

ORBIT PLATFORM DOCUMENTATION

Offline Multimodal RAG Engine · Full Technical Specification

1. EXECUTIVE OVERVIEW

Orbit is a 100% offline Multimodal RAG Platform providing enterprise privacy, air-gapped document intelligence, and zero cloud

2. KEY FEATURES

- Support for PDF, Word, PowerPoint, Excel, Images, Audio, Video, SQLite DBs.
- 3-Column Orbit Dashboard UI.
- Hybrid BM25 Keyword + Dense Vector Search.
- Agentic Multi-Hop Query Decomposition & GraphRAG.
- Self-RAG Reflective Critic Agent.

3. TECHNICAL STACK

- Frontend: Streamlit 1.39+ with CSS Glassmorphism
- Local LLM: Ollama llama3.2:3b
- Embedding Model: nomic-embed-text (768-dim)
- Vector Store: ChromaDB Persistent HNSW Client
- Search Engine: Rank-BM25 (BM25Okapi)
- Transcription: faster-whisper (CTranslate2)
- Executable: PyInstaller 6.22 Standalone EXE

4. ALGORITHMS USED

- Reciprocal Rank Fusion (RRF): $\text{Score} = 1 / (60 + \text{vec_rank}) + 1 / (60 + \text{bm25_rank})$
- Okapi BM25 Keyword Scoring ($k1=1.5$, $b=0.75$)
- Dense Vector Cosine Similarity
- Overlapping Recursive Character Window Chunking

5. FUTURE IMPLEMENTATIONS

- Local LLaVA Multimodal Vision LLM Integration
- Cross-Encoder BGE-Reranker-Large
- Electron.js Native Desktop App Packaging
- Sub-300ms WebRTC Real-Time Voice Mode